Jitin Kumar

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RESEARCH INTERESTS

Applied machine learning and NLP for climate action and education, with a focus on predictive modelling and AI systems that are explainable and personalised.

EDUCATION

B. Tech, Computer Science Engineering (AI) - *CSJM University (UIET), Kanpur* Aug 2024 - Jun 2028

- CGPA 8.71 / 10; consistently in the top 15% of the B. Tech CSE (AI)
- Relevant coursework: Machine Learning, Computer Vision, Data Structures, Object-Oriented Programming, Operating Systems

RESEARCH AND APPLIED EXPERIENCE

Green Skills and Applied AI for Climate Action Intern - *1M1B* Jun 2026 - Aug 2026

- Finished a 70+ hour programme on using AI for climate action and green skills, with Microsoft Elevate, MeitY Startup Hub and UP Pragya
- Built and deployed SkillGreen, a full-stack app that predicts ESG career readiness (FastAPI, scikit-learn, React); now adding resume parsing, model explainability and skill-gap analysis

Data Science and Analytics Intern - *Tamizhan Skills (Remote)* Jun 2025 - Jul 2025

- Worked through the full data workflow (loading, cleaning, exploratory analysis) on 5+ datasets using Pandas, Matplotlib and Seaborn
- Cut dataset inconsistencies by ~25% by catching outliers and filling in missing values, which made the downstream models more dependable
- Brought data-handling time down by ~40% by swapping loops for vectorised NumPy operations
- Built 3 interactive Plotly dashboards, so stakeholders needed fewer one-off report requests

Machine Learning Intern - *TechKriti, IIT Kanpur* Jan 2025 - Feb 2025

- Built 2 supervised models (an NLP classifier and a regression model), handling training, evaluation and refinement
- Shortened training time by ~20% with feature selection and faster Scikit-learn preprocessing
- Placed in the top 3 intern submissions at the internal showcase and presented the results to an IIT Kanpur faculty panel

AWS APAC Solutions Architecture - *Forage Virtual Program* Jan 2025

- Designed a scalable AWS Elastic Beanstalk hosting setup for a simulated client brief, and wrote up the recommendation in plain language for a non-technical audience

PROJECTS

SkillGreen: ESG Career-Readiness Predictor GitHub | Live 2026

- Predicts how ready someone is for an ESG (Environmental, Social, Governance) career from their profile, using a scikit-learn model, a FastAPI + Pydantic backend and a React frontend; built during the 1M1B internship
- Deployed with the frontend on Vercel and the backend and database on Render, using Docker; resume parsing, explainability and a skill-gap engine are being added
- Stack: Python, FastAPI, Pydantic, Scikit-learn, React, Vite, PostgreSQL, Docker

RiskLens: Insurance Premium Risk Predictor GitHub 2026

- Classifies a user's profile into a Low / Medium / High insurance premium risk category, with feature engineering built into the model's input pipeline and served through FastAPI
- Stack: Python, FastAPI, Feature Engineering

ModelServe: FastAPI Serving Layer for ML Models GitHub 2026

- Built a FastAPI service that exposes machine learning models as REST APIs, with endpoints for preprocessing, prediction and evaluation, plus auto-generated API docs
- Stack: Python, FastAPI, REST APIs, Jupyter

Student Performance Analytics Dashboard GitHub 2025

- Combined 3 data sources (grades, attendance, login activity) into one pipeline that feeds a dashboard for faculty to spot at-risk students
- Trained a Random Forest classifier (87% accuracy) to flag low-performing students before the end of term; faculty adopted the per-student improvement suggestions for early intervention
- Stack: Python, Pandas, Scikit-learn, Matplotlib, Jupyter

TECHNICAL SKILLS

Languages Python, C/C++
ML / AI: Linear Regression, Random Forest, Feature Engineering, NLP, Predictive Modelling, Computer Vision (coursework)
Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly
Tools: Jupyter, Git, VS Code, Docker, FastAPI, Pydantic, React, AWS Elastic Beanstalk

CERTIFICATIONS AND ACTIVITIES

- Machine Learning Certification, TechKriti, IIT Kanpur (Feb 2025)
- Data Science and Analytics Certification, Tamizhan Skills (Jul 2025)
- Machine Learning with Python, FreeCodeCamp (Dec 2025)
- Green Skills and Applied AI for Climate Action Internship, 1M1B (Aug 2026)
- National Service Scheme (NSS): active volunteer